SOURCE SECTION 1

Simulated Teaching Hospital

Medical Record Copy Format / v1.1 / 2026-09-15 / Pending clinical expert review

Name: Wang Jianhua (fictional) Sex: Male Age: 56 years Department: Gastroenterology Medical record number: SIM-ASC-20260901

Case Design and Teaching Objectives for Instructors

Version and Scope of Use

v1.1 / 2026-09-15 / Pending clinical expert review. This booklet contains answers and a predefined clinical course. The patient is Wang Jianhua, a 56-year-old man. The central clinical course involves worsening ascites due to cirrhosis, spontaneous bacterial peritonitis (SBP), reversible acute kidney injury (AKI), and improvement after treatment.

Overall Objective

As information is presented in stages, senior residents should independently formulate a problem representation, establish priorities, select investigations, interpret results, initiate treatment, and reassess, while explaining the evidence that triggers changes to the management plan.

Teaching Objectives and Assessment Evidence

Objective	Observable Performance	Assessment Evidence
O1 Problem representation	Summarize the liver disease background, changes in ascites, and warning signs in one sentence, prioritizing evaluation for infection and inadequate perfusion.	Initial progress note / oral handover
O2 Etiology of ascites	Calculate SAAG = 25 - 7 = 18 g/L, and use ascitic fluid protein and clinical findings to differentiate cardiac, malignant, tuberculous, and other causes.	Investigation selection and result interpretation
O3 Occult infection	Perform paracentesis promptly even without fever; a PMN count of 640 triggers treatment for SBP without waiting for culture results.	Timing of actions and rationale
O4 Dynamic recognition of AKI	Recognize the short-term change from 118 to 150; do not infer the time of onset solely from a baseline value more than 7 days old.	Trend interpretation and diagnostic revision
O5 Individualized treatment	Distinguish the infection phase, renal function recovery phase, and ascites control phase; explain when to stop or resume diuretics and the indications for albumin.	Medical orders and reassessment plan
O6 Treatment response	Recognize the 81.25% decrease in PMN count together with improvement in symptoms and creatinine; continue treatment according to antimicrobial susceptibility results and avoid escalation without supporting evidence.	Ward round notes
O7 Escalation and differential diagnosis	Reassess for hepatorenal syndrome (HRS) and other causes if kidney injury persists; investigate secondary peritonitis if treatment fails; verify the definition of refractory ascites.	Branch-based reasoning
O8 Continuity of care	Clearly specify secondary prophylaxis for SBP, nutrition, treatment of the underlying cause, monitoring, and transplant evaluation, with the patient able to repeat the plan back.	Discharge plan and communication

Assessment focuses on evidence, timing, and the ability to revise decisions. Scoring weights, the passing threshold, and whether any critical error constitutes an automatic failure have not been established and are pending review and finalization by experts and the teaching team.

SIM-ASC-20260901 | Simulated teaching case · Not a real medical document

SOURCE SECTION 2

Simulated Teaching Hospital

Medical record copy format / v1.1 / 2026-09-15 / Pending clinical expert review

Patient information

Name	Sex	Age	Department	Medical record number
Wang Jianhua (fictional)	Male	56 years	Gastroenterology	SIM-ASC-20260901

Case Answer Key · Key Clinical Reasoning

1. Why investigate infection first?

The patient was hospitalized for recently worsening ascites, with abdominal discomfort and renal dysfunction. The absence of high fever and an elevated blood WBC count does not exclude spontaneous bacterial peritonitis (SBP). The absolute ascitic fluid polymorphonuclear neutrophil (PMN) count, rather than a positive culture, is the key to timely action.

2. Why not directly equate this with hepatorenal syndrome?

A creatinine increase of 32 μmol/L within 20 hours confirms acute kidney injury (AKI). Recent inadequate intake, a self-increased diuretic dose, and infection may coexist. Urine sediment examination and ultrasonography are used to investigate alternative causes, but normal results alone cannot confirm hepatorenal syndrome (HRS). Given the rapid recovery after stopping diuretics and treating the infection, HRS-AKI is not documented as a confirmed final diagnosis. [S5]

3. Why is treatment staged?

On the first day, control infection, withhold diuretics, and assess circulatory status. After infection and renal function improve, perform paracentesis, then resume diuretics at low doses. Albumin dosing for SBP is calculated using estimated dry weight; albumin dosing for paracentesis is calculated using the volume drained. These must not be conflated into “give albumin whenever the albumin level is low.” [S3,S4]

4. Why not use TIPS at present?

Ascites has not yet been shown to be refractory after adequate, tolerable treatment and verification of adherence; it has improved during this admission. A transjugular intrahepatic portosystemic shunt (TIPS) is an option for recurrent refractory ascites, requiring assessment of cardiopulmonary and liver function, infection, and encephalopathy risk. Transplant evaluation can begin after decompensation, without waiting for all treatments to fail. [S2,S6]

5. Why is follow-up still necessary after discharge?

AKI has resolved, but hyponatremia, hypoalbuminemia, portal hypertension, and chronic liver disease persist. After SBP, secondary prophylaxis, antiviral therapy, nutrition, and management of liver cancer and varices need to be discussed. “Improved” cannot substitute for a specific reassessment and safety plan.

Guideline update: ADQI-ICA 2024 recommends against routinely requiring 48 hours of albumin administration as a prerequisite for diagnosing HRS-AKI; specific volume management depends on clinical assessment. Debate in this area continues, and scoring is not tied to a single rigid algorithm.

SIM-ASC-20260901 | Simulated teaching case · Not a real medical document

SOURCE SECTION 3

Simulated Teaching Hospital

Medical Record Copy Format / v1.1 / 2026-09-15 / Pending Clinical Expert Review

Name: Wang Jianhua (fictional) | Male | Age: 56 years | Department: Gastroenterology | Medical Record No.: SIM-ASC-20260901

Information Disclosure and Role Guidelines for Five Scenarios

Scenario-Based Information Release and Learner Tasks

Scenario	Information Release	Learner Tasks
Initial assessment	09-01 09:20–11:00; three pages of admission history and physical examination, prior outpatient summary, ASC-L01, ASC-L02, ASC-L03, urine section of ASC-L04, ASC-US01, ASC-ECG01	Complete the problem representation and differential diagnosis; propose abdominal paracentesis and necessary investigations; propose initial volume/diuretic management.
Management and handover	09-01 11:45–18:00; ASC-A01, first-day treatment response	Calculate SAAG and PMN count; manage SBP and communicate monitoring requirements; distinguish indications for albumin.
Ward rounds	Progress by date from 09-02 to 09-08; daily laboratory results are unlocked at their report times, preliminary and final culture reports are unlocked separately, and procedure records become visible after performance.	Confirm AKI and establish the differential diagnosis; adjust medications and the volume management plan; reassess treatment response and decide on therapeutic paracentesis and resumption of diuretics.
Discharge documentation	09-09; all information obtained previously and pre-discharge investigations	Document accurate diagnoses and outcomes; specify discharge medications and follow-up; communicate risks.
Feedback	Upon release by the instructor or after the exercise ends; instructor teaching manual and complete medical record	Compare the evidence available at the time of decision-making; explain reasons for omissions or adjustments; reflect on acceptable alternatives.

Patient Role Guidelines

The patient may describe symptoms, recently added medications, diet, urine output, and known medical history, but should not spontaneously use technical diagnostic terms such as “SBP,” “SAAG,” or “HRS.” When asked, provide information truthfully; do not conceal established facts to increase the difficulty.

Supervising Clinician Role Guidelines

In practice mode, graded hints about “what evidence is still missing” may be provided on request. In examination mode, do not provide the standard diagnosis or future investigation results in advance. Feedback should evaluate performance solely on the basis of information available at the time.

Boundaries of Data Use

The JSON contains the complete facts and clinical course for instructors and is intended only for future integration; this does not mean that server-side access controls, automated scoring, or causal simulation of the clinical condition have been implemented. The current clinical course is a reference pathway and does not imply that any learner action will produce the same outcome.

SIM-ASC-20260901 | Simulated Teaching Case · Not a Real Medical Document

Simulation Teaching Hospital

Medical record reproduction format / v1.1 / 2026-09-15 / Pending clinical expert review

Name: Wang Jianhua (fictional) Male 56 years old Department: Gastroenterology Medical record number: SIM-ASC-20260901

Advanced Branches and Acceptable Alternatives

Hypothetical branches separate from the main course

Trigger (did not occur in the main course)	Reasoning to demonstrate	Separation from the main course
Insufficient PMN decline at 48 hours or worsening abdominal pain	Verify antimicrobial coverage, resistance, and sampling; consider secondary peritonitis, repeat imaging, and seek a surgical consultation as appropriate.	Do not incorporate polymicrobial findings or perforation into this patient's facts.
Persistently rising creatinine / hypoperfusion	Assess infection control, volume status, nephrotoxic drugs, and renal parenchymal disease; discuss HRS treatment only if the criteria are met.	Failure to improve after stopping diuretics alone must not be treated as evidence of HRS.
Recurrent large-volume ascites with failure of, or intolerance to, adequate treatment	Verify sodium restriction and medication adherence; assess for refractory ascites and discuss repeated paracentesis / TIPS / transplantation.	Refractory ascites has not been diagnosed in the current case.
New-onset melena or altered mental status	Prioritize stabilization, identify precipitating factors, and assess according to the relevant complication pathway.	Do not arbitrarily add these complications to increase case complexity.

Acceptable Variations

Antimicrobial selection may vary according to local resistance patterns, community versus hospital onset, and prior exposure; regional protocols may also differ in albumin dosing strategies and the timing of resuming diuretics. The rationale must be explained and monitoring performed; an approach must not be marked incorrect solely because it differs from this reference prescription.

Medication List and Actual Medication Use

Only data on medications actually used in this case are provided. Terlipressin, lactulose, rifaximin, and similar drugs belong to branches that may be discussed and are not documented as having been used in this case. The entecavir dosing interval was adjusted once at the peak of AKI; this must be kept consistent with the clinical course and medication orders.

SIM-ASC-20260901 | Simulated teaching case · Not a real medical document

Simulated Teaching Hospital

Medical record reproduction format / v1.1 / 2026-09-15 / Pending clinical expert review

Name: Wang Jianhua (fictional) Male 56 years old Department: Gastroenterology Medical record no.: SIM-ASC-20260901

Expert Review Form · Not Yet Reviewed

Expert review checklist

Review area	Key points to check	Comments / Revisions
Hepatology and infection	History, evidence for SBP, culture results, and appropriateness of treatment duration and secondary prophylaxis	_____
Renal function and volume status	AKI time window/staging, volume assessment, differential diagnosis of HRS, restarting diuretics, and medication dose adjustments	_____
Pharmacy	Local formulations, administration methods, albumin dose calculation, ciprofloxacin risks, and renal dose adjustment of entecavir	_____
Procedures and nursing	Informed consent, safe ultrasound window, volume drained, monitoring, and bleeding risk assessment	_____
Imaging and laboratory testing	Plausibility of narrative reports and numerical values; suitability of synthetic images for display, which must not be used as a diagnostic gold standard	_____
Teaching	Difficulty of learning objectives, chronological ordering of materials, acceptable alternatives, and feedback methods	_____

Key Omissions (Pending Review and Finalization)

Failure to arrange timely paracentesis; delaying SBP management while awaiting culture results; indiscriminately increasing diuretic therapy during AKI; directly diagnosing all AKI in cirrhosis as HRS; failure to reassess despite nonresponse to treatment; absence of a post-discharge monitoring and secondary prophylaxis plan.

Review Conclusion

□ Major revisions required □ Minor revisions required □ Suitable for limited teaching use □ Other: _____

Reviewing expert: _____ Specialty: _____ Date: _____

The above fields are blank sections for manual review and do not indicate that an expert has signed off. Automated checks cover only numerical values, chronology, and formatting and cannot replace medical review.

SIM-ASC-20260901 | Simulated teaching case · Not a real medical document

SOURCE SECTION 6

Simulated Teaching Hospital

Medical Record Copy Format / v1.1 / 2026-09-15 / Pending clinical expert review

Patient information

Name	Sex	Age	Department	Medical record number
Wang Jianhua (fictional)	Male	56 years	Gastroenterology	SIM-ASC-20260901

Sources and Version Notes

S1 · Knowledge framework from the cirrhosis chapter in Internal Medicine

The user did not specify an edition, and the original textbook text was not obtained; no page numbers or verbatim quotations have been fabricated. An expert will supplement the textbook cross-references.

S2 · Chinese Society of Hepatology, Chinese Medical Association: Guidelines on the Diagnosis and Treatment of Ascites in Cirrhosis (2023 edition)

Chinese Journal of Hepatology 2023;31:813-826. Used as the domestic framework for the diagnosis and treatment of ascites; access to the full text in the Chinese-language journal is restricted.

https://pubmed.ncbi.nlm.nih.gov/37723063/

S3 · AASLD: Ascites, SBP and HRS Practice Guidance (2021)

Supports initial management of SBP, albumin use, and long-term management; the kidney injury section incorporates updates from S5.

https://doi.org/10.1002/hep.31884

S4 · BSG/BASL: Guidelines on the management of ascites in cirrhosis (2021)

Supports ascitic fluid testing, albumin at 8 g/L after paracentesis, and alternative options for secondary prophylaxis of SBP.

https://doi.org/10.1136/gutjnl-2020-321790

S5 · ADQI-ICA: Acute kidney injury in patients with cirrhosis (2024)

Supports AKI recognition, volume status assessment, and differential diagnosis of HRS-AKI; opposes making routine albumin infusion for 48 hours a mandatory diagnostic requirement in all patients.

https://pubmed.ncbi.nlm.nih.gov/38527522/

Search verification date: 2026-09-15. No claim is made that all guidelines available through that date were exhaustively covered. Recommendations differ across regions; expert-determined teaching standards and local protocols take precedence.

SIM-ASC-20260901 | Simulated teaching case · Not a real medical document

SOURCE SECTION 7

Simulated Teaching Hospital

Medical record copy format / v1.1 / 2026-09-15 / Pending clinical expert review

Name: Wang Jianhua (fictional) Sex: Male Age: 56 years Department: Gastroenterology Medical record number: SIM-ASC-20260901

Supplementary Evidence and Calculation Checks

S6 · Chinese Guidelines for Clinical Diagnosis, Treatment, and Management of Cirrhosis (2025)

Cross-check the framework for nutrition, etiology, and long-term follow-up in decompensated cirrhosis.

https://doi.org/10.14218/JCTH.2025.00517

S7 · KASL clinical practice guidelines: Ascites and related complications (2026)

A 2026 update identified through a supplementary search, provided for experts to compare regional differences; not a local prescribing standard for China.

https://pmc.ncbi.nlm.nih.gov/articles/PMC13430399/

S8 · DailyMed: Entecavir tablets prescribing information

Basis for 1 mg once daily in adults with decompensated disease, administration on an empty stomach, and dose adjustment for renal function; local prescribing information must be rechecked before actual use.

https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=1a905283-0057-a7dd-e063-6394a90a1909

Calculation Checks

Item	Calculation / Explanation
SAAG	25-7=18 g/L (paired samples before albumin infusion)
Initial PMN	1000×64%=640 cells/μL
48-hour PMN change	(640-120)/640=81.25% decrease
Evidence of AKI	150-118=32 μmol/L / approximately 20 hours; 150/78=1.923-fold
Albumin for SBP	68×1.5=102 g; 68×1=68 g
Albumin after paracentesis	6 L×8 g/L=48 g; total across three administrations: 218 g
Albumin volume	20%=0.2 g/mL; 510+340+240=1090 mL (over 3 days)

This teaching case is an original synthetic case and does not copy cases from guidelines or original reports. The ultrasound image was created using a built-in image-generation tool; see the asset notes for the prompt and asset sources. The electrocardiogram was drawn using a local program.

SIM-ASC-20260901 | Simulated teaching case · Not a real medical document